

Jinsung Park

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Experience

Software Development Engineer

Jul 2024 – Present

AWS Galaxi DWaaS (Data Warehouse as a Service)

- Automated Redshift provisioning state machine with federated URL and Quicksight role creation.
- Reconstructed the company's bandwidth critical operation for sellers and developers from first principles, utilizing unified authentication source and GraphQL schema to expose application layer data.
- Maintained AWS Galaxi's hexagonal architecture, streamlining access by migrating Isengard and Bindle permissions into the new Galaxi Portal.
- Drafted a 21-page Operational Reliance Review to support new feature rollouts, enabling the SUDS team to scale its low-latency applications with higher reliability.
- Maintained 99.8% uptime across NOVA, DWaaS, Airflow, GPSS, and Bedrock by triaging 132+ high severity tickets.

Research Assistant

Nov 2021 – June 2024

UCSC Vertical, Architecture, Memories, and Algorithms (VAMA)

- Extended Python compiler for FIRRTL (HDL intermediate representation for RTL), adding 42 operations emitting C++ code.
- Produced a test bench to find 11+ bugs in a codebase implementing UInt and SInt types.
- Added 42 low-level critical operations into the Python library defined in FIRRTL to generate C++ code.
- Utilized Hypothesis and fuzzy algorithms to induce smart predictions in the model for better accuracy.

Projects

tryai.cinna.dev | *Rails, Ruby, PostgreSQL, Qdrant, Embeddings, RAG*

- Designed and implemented a complete infrastructure from scratch via Figma, Excalidraw, and Rails.
- Architected Rails for horizontal scalability, designed for concurrent users with redundancy targeting 99.5% uptime.

Digital F and A Accumulator | *STM32, Embedded C, Zynq 7010, Verilog*

- Initially built waveform generator in CPU with F410RB and MCP4921 DAC then upgraded to FPGA with Zybo-z7.
- Designed FIR module to reduce noise and CC1101 Driver handle communication with RF device
- Optimized the timing of a custom software-defined waveform generator by utilizing a precisely tuned Timer Interrupt Service Routine (ISR) to stream data over SPI.

Chisel AI Accelerator | *Scala, Chisel, TPU, Verilog*

- Designed and implemented a DNN accelerator generator, achieving a processing speed of 5 TOPS during testing, enhancing performance metrics for real-time ML applications.
- Built and validated a Chisel-based hardware simulation pipeline for systolic arrays, implementing dot-product operations with clean, testable, and extensible code.

Optical Character Recognition for Trivia Games | *Java*

- Designed an optical character recognizing software for reading characters on screen and scraping webs.
- Parses questions through OCR provided by trivia games and provides answers based on web results.
- Correctly answers trivia questions with 3 choices within 5 seconds with 93% accuracy.

Opensource Contributions

Xilinx/embeddedsw

- Contributed a bug fix to Xilinx open source embeddedsw repo to fix timer configuration in AXI timer so that initialization resets PWM and the two interface counters to prevent stale memory from corrupting new data. (PR #383)

Skills

Rust, Python, Verilog, Scala, C/C++, RISC-V, ESP32C3, FPGA, Xilinx, RTOS, Espressif, Vivado, STM32CubeIDE, Vitis, Kicad

Education

B.S. Computer Engineering + Economics (3.97 GPA)

2020 – 2024